

MATH 375: Probability Theory

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Lecture 5

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1 Lecture 5: Multiplication, Total Probability, and Independence

1.1 The multiplication rule

For events A and B , provided $P(A) > 0$,

$$P(A \cap B) = P(A)P(B | A).$$

The rule extends to several events. For example,

$$P(E_1 \cap E_2 \cap E_3) = P(E_1) P(E_2 | E_1) P(E_3 | E_1 \cap E_2).$$

In general, each new factor is conditioned on all the events already listed before it.

1.2 Example: separating four aces

A shuffled deck is divided into four labeled piles of 13 cards. What is the probability that every pile contains exactly one ace?

Imagine placing the four aces into piles one at a time. The first ace can go into any pile. For the other three aces to go into different piles, their choices must differ from the earlier ones:

$$\begin{aligned} P(\text{one ace in each pile}) &= 1 \cdot \frac{3}{4} \cdot \frac{2}{4} \cdot \frac{1}{4} \\ &= \boxed{\frac{3}{32}}. \end{aligned}$$

1.3 The law of total probability

Suppose $F_1, \dots, F_n$ are mutually exclusive events whose union is the whole sample space. They form a **partition**: exactly one of them occurs.

Any event E can be split into disjoint pieces:

$$E = (E \cap F_1) \cup \dots \cup (E \cap F_n).$$

Adding the probabilities of those pieces gives

$$P(E) = \sum_{i=1}^n P(F_i)P(E | F_i).$$

1.4 Example: choosing a coin

A box contains two type A coins and one type B coin. A type A coin lands heads with probability $\frac{1}{4}$, while a type B coin lands heads with probability $\frac{3}{4}$. One coin is selected uniformly and flipped. Given that it lands heads, what is the probability that it was type A?

Let A mean “type A was selected,” B mean “type B was selected,” and H mean “the flip is heads.” Then

$$P(A) = \frac{2}{3}, \quad P(B) = \frac{1}{3},$$

and

$$P(H | A) = \frac{1}{4}, \quad P(H | B) = \frac{3}{4}.$$

First, total probability gives

$$\begin{aligned} P(H) &= P(A)P(H | A) + P(B)P(H | B) \\ &= \frac{2}{3} \cdot \frac{1}{4} + \frac{1}{3} \cdot \frac{3}{4} \\ &= \frac{1}{6} + \frac{1}{4} = \frac{5}{12}. \end{aligned}$$

Now apply Bayes’ formula:

$$P(A | H) = \frac{P(A)P(H | A)}{P(H)} = \frac{1/6}{5/12} = \frac{2}{5}.$$

1.5 Example: at least one girl

Assume each child is independently equally likely to be a boy or a girl. For a two-child family, use ordered outcomes:

$$S = \{BB, BG, GB, GG\}.$$

Given that *at least one child is a girl*, the remaining equally likely possibilities are BG , GB , and GG . In only one of these are both children girls. Therefore,

$$P(\text{both girls} | \text{at least one girl}) = \frac{1}{3}.$$

The wording of the given information determines which outcomes remain in the sample space.

1.6 Example: a girl born on Tuesday

Assume birth sex and weekday are independent and equally likely. Suppose we are told that *at least one of two children is a girl born on Tuesday*. What is the probability that both children are girls?

For one child, the probability of *not* being a girl born on Tuesday is 13/14. Thus,

$$P(\text{at least one girl born Tuesday}) = 1 - \left(\frac{13}{14}\right)^2 = \frac{27}{196}.$$

For both children to be girls and at least one to have been born on Tuesday, consider their two birth weekdays. There are 7^2 pairs of weekdays, of which 6^2 have no Tuesday:

$$P(\text{both girls and at least one Tuesday girl}) = \frac{1}{4} \left(1 - \frac{6^2}{7^2}\right) = \frac{13}{196}.$$

Therefore,

$P(\text{both girls} \mid \text{at least one Tuesday girl}) = \frac{13/196}{27/196} = \frac{13}{27}.$

1.7 Independent events

Events A and B are **independent** if

$P(A \cap B) = P(A)P(B).$

If $P(A) > 0$, this is equivalent to

$$P(B \mid A) = P(B).$$

Knowing whether A occurred does not change the probability of B .

Example: two dice. Let F mean “the first die shows 4,” so $P(F) = 1/6$.

Let E_5 mean “the sum is 5.” There are five ways to get a sum of 5, and only $(4, 1)$ also belongs to F . Thus,

$$P(E_5) = \frac{5}{36}, \quad P(E_5 \cap F) = \frac{1}{36},$$

but

$$P(E_5)P(F) = \frac{5}{216} \neq \frac{1}{36}.$$

So E_5 and F are **dependent**.

Now let E_7 mean “the sum is 7.” There are six ways to get a sum of 7, and only $(4, 3)$ also belongs to F :

$$P(E_7) = \frac{6}{36} = \frac{1}{6}, \quad P(E_7 \cap F) = \frac{1}{36}.$$

This time,

$$P(E_7)P(F) = \frac{1}{6} \cdot \frac{1}{6} = \frac{1}{36},$$

so E_7 and F are **independent**.